

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

Author:

Tahir JAVAID

Supervisor:

Prof. Guoming CHEN

Co-supervisor:

Prof. Mingshui CHEN

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*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

Chapter 1

Probing four lepton final-state for Higgs Cross section measurement

1.1 Introduction

In 2012, CMS and ATLAS collaborations announced the discovery of the existence of new boson [Reference20, Reference21] with mass around 125 GeV. The comprehensive properties of the newly discovered boson have been studied by both CMS and ATLAS experiments [Reference22, Reference21a, Reference21b, Reference21c] and are found to be compatible with SM Higgs boson.

In this chapter, measurements of the integrated and differential fiducial cross sections for the Higgs boson production in the $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) with the proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale. Any deviation of the integrated or differential distributions from SM expectations would provide a direct hint to BSM contributions. Nearly similar measurements have already been reported by ATLAS experiment in $H \rightarrow \gamma\gamma$ [ATLAS:Hgg_8TeV, ATLAS:Hgg_8TeV_13TeV], $H \rightarrow 4\ell$ [ATLAS:H4l_8TeV, ATLAS:H4l_8TeV_13TeV] decay channels, combination of both channels [ATLAS:H4l_Hgg_8TeV, ATLAS:H4l_Hgg_8TeV_13TeV] and by CMS experiment in $H \rightarrow \gamma\gamma$ decay channel [CMS:Hgg_8TeV].

The integrated fiducial cross section measurements are performed using collisions data corresponding to integrated luminosity of 5.1 fb^{-1} at $\sqrt{s} = 7$ TeV and 19.7 fb^{-1} at $\sqrt{s} = 8$ TeV, recorded with the CMS detector at LHC. The measurement of the integrated ratio of the fiducial cross sections at 7 TeV and 8 TeV is also logged. The differential fiducial cross sections are measured with 8 TeV data only, because insufficient statistics of 7 TeV data. To minimize the dependence of the results on the underlying models of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial phase space closely matching with the experimental acceptance as shown in Figure 1.1. The remaining model dependence is extracted by repeating the measurements for SM and other exotic models and reported as a distinct systematic uncertainty. All measurements includes the corrections for the finite detection efficiency and resolution effects. As $\sigma \times BR$ strongly depends on the Higgs mass, the measurements have been extracted assuming $m_H = 125.0$ GeV, as measured using the $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay channels by the CMS experiment [Reference39].

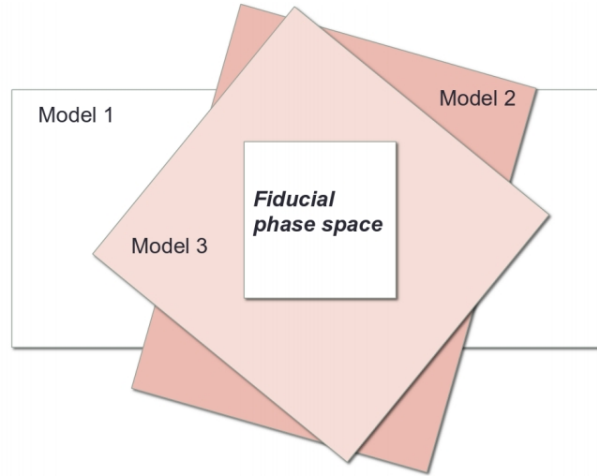


FIGURE 1.1: Fiducial phase definition among different models.

The differential fiducial cross sections are measured as a function of kinematic properties of the four-lepton system and the leading jet, as well as a function of the jet multiplicity. The results are reported for several kinematic observables which are sensitive to the Higgs-boson production mechanisms: transverse momentum and the rapidity of the four-lepton system, transverse momentum of the leading jet, as well as distance in rapidity between the Higgs boson candidate and the leading jet. The results are also reported for kinematic observables which are sensitive to the Higgs boson decay mechanisms: the invariant masses of the two lepton pairs and five helicity angles formed by the leptons in the Collins-Soper frame [Reference40].

The chapter is structured as follows. The datasets and simulated samples used for the analysis are introduced in Section ???. The analysis strategy is described in Section ??, where fiducial phase space definition, event selection and method for extracting integrated and differential fiducial cross sections are explained. In Section ?? details of systematic uncertainties affecting the measurement are explained. In Section ?? results are compared with SM theoretical predictions.